

Cyclic State Detection via Cryptographic Signatures vs. Scalar Violation Counts

Cryptographic Match: $H(S_{t+2}) = H(S_t) \rightarrow$ Infinite Cycle Detected!



The Naive Scalar Count Trap (Blind to State Identity)

- Scalar Monitoring: Counts total violations (15 $\rightarrow$ 15 $\rightarrow$ 15).
- The Illusion: Optimizer assumes continuous search progress or stalls because violation count does not decrease.
- The Hazard: Cannot distinguish productive exploration along a Pareto boundary from cyclic thrashing between two inverted failure modes (setup timing vs. hold / routing congestion).

Cryptographic Cycle Detection (The Architectural Antidote)

- State Fingerprint: Hashes RTL AST, SDC constraints, and UPF intent.
- Exact Cycle Check: Evaluates $H(S_t) == H(S_{t+2})$ equality.
- Deterministic Halt: Identifies that state S_{t+2} exactly repeats S_t despite identical violation counts.
- Recovery Action: Breaks the infinite loop, releases EDA licenses, rolls back to last trusted checkpoint, and requests strategy change.